

# HDOC – Day 24

## C Programming Assignment

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**Batch 16**

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### **Problem Requirements**

Display a menu of five increasing-pattern options. Use a switch statement to select the user's choice. Each pattern must be implemented in a separate parameterless function with no return value (void). For each activity, the algorithm and test cases are provided below, followed by one complete C program.

#### **1. Increasing right-angled triangle**

##### **Algorithm**

1. Start.
2. Input number of rows n.
3. For i = 1 to n, print i asterisks using an inner loop.
4. Move to the next line after each row.
5. Stop.

| Test Input | Expected Output                | Result |
|------------|--------------------------------|--------|
| n=4        | <pre>* * * * * * * * * *</pre> | Pass   |
| n=1        | <pre>*</pre>                   | Pass   |

#### **2. Increasing row-number triangle**

##### **Algorithm**

1. Start.
2. Input number of rows n.

3. For each row  $i$  from 1 to  $n$ , print the value  $i$  exactly  $i$  times.
4. Move to the next line after each row.
5. Stop.

| Test Input | Expected Output              | Result |
|------------|------------------------------|--------|
| $n=4$      | 1<br>2 2<br>3 3 3<br>4 4 4 4 | Pass   |
| $n=2$      | 1<br>2 2                     | Pass   |

### 3. Increasing natural-number pattern

#### Algorithm

1. Start.
2. Input number of rows  $n$ .
3. For each row  $i$  from 1 to  $n$ , print numbers  $j = 1$  through  $i$ .
4. Move to the next line after each row.
5. Stop.

| Test Input | Expected Output              | Result |
|------------|------------------------------|--------|
| $n=4$      | 1<br>1 2<br>1 2 3<br>1 2 3 4 | Pass   |
| $n=3$      | 1<br>1 2<br>1 2 3            | Pass   |

### 4. Consecutive-number triangle

#### Algorithm 1.

Start.

2. Input number of rows  $n$  and set  $num = 1$ .
3. For each row  $i$  from 1 to  $n$ , print  $i$  values of  $num$ ; increment  $num$  after every print.

4. Move to the next line after each row.
5. Stop.

| Test Input | Expected Output               | Result |
|------------|-------------------------------|--------|
| n=4        | 1<br>2 3<br>4 5 6<br>7 8 9 10 | Pass   |
| n=3        | 1<br>2 3<br>4 5 6             | Pass   |

## 5. Increasing alphabet pattern

### Algorithm

1. Start.
2. Input number of rows n.
3. If n is outside 1 to 26, display an error and return.
4. For each row i from 1 to n, print characters from A through A+i-1.
5. Move to the next line after each row.
6. Stop.

| Test Input | Expected Output                         | Result |
|------------|-----------------------------------------|--------|
| n=4        | A<br>A B<br>A B C<br>A B C D            | Pass   |
| n=26       | Rows increase from A up to A ... Z      | Pass   |
| n=27       | Invalid input. Enter rows from 1 to 26. | Pass   |

### Complete C Program #include <stdio.h>

```
void starTriangle(void); void
rowNumberTriangle(void); void
naturalNumberPattern(void); void
```

```

consecutiveNumberTriangle(void); void
alphabetPattern(void); int main(void)
{
int choice;
printf("INCREASING PATTERNS MENU\n");
printf("1. Increasing right-angled triangle\n");
printf("2. Increasing row-number triangle\n");
printf("3. Increasing natural-number pattern\n");
printf("4. Consecutive-number triangle\n");
printf("5. Increasing alphabet pattern\n");
printf("Enter your choice: "); scanf("%d", &choice);
switch (choice) { case 1: starTriangle(); break; case
2: rowNumberTriangle(); break; case 3:
naturalNumberPattern(); break; case 4:
consecutiveNumberTriangle(); break; case 5:
alphabetPattern(); break;
default: printf("Invalid choice.\n");
}
return 0;
}
void starTriangle(void)
{
int n, i, j;
printf("Enter number of rows: "); scanf("%d",
&n);
if (n <= 0) { printf("Invalid input.\n"); return; }
for (i = 1; i <= n; i++) { for (j = 1;
j <= i; j++) printf("* ");
printf("\n");
}
}
void rowNumberTriangle(void)
{
int n, i, j;
printf("Enter number of rows: "); scanf("%d",
&n);

```

```

if (n <= 0) { printf("Invalid input.\n"); return; }
for (i = 1; i <= n; i++) {
for (j = 1; j <= i; j++) printf("%d ", i);
printf("\n");
}
}

void naturalNumberPattern(void)
{
int n, i, j;
printf("Enter number of rows: "); scanf("%d",
&n);
if (n <= 0) { printf("Invalid input.\n"); return; }
for (i = 1; i <= n; i++) { for (j = 1; j <=
i; j++) printf("%d ", j); printf("\n");
}
}

void consecutiveNumberTriangle(void)
{
int n, i, j, num = 1; printf("Enter
number of rows: "); scanf("%d",
&n);
if (n <= 0) { printf("Invalid input.\n"); return; }
for (i = 1; i <= n; i++) {
for (j = 1; j <= i; j++) printf("%d ", num++);
printf("\n");
}
}

void alphabetPattern(void)
{
int n, i, j;
printf("Enter number of rows (1-26): ");
scanf("%d", &n);
if (n < 1 || n > 26) {
printf("Invalid input. Enter rows from 1 to 26.\n"); return;
}
}

```

```

for (i = 1; i <= n; i++) {
for (j = 0; j < i; j++) printf("%c ", 'A' + j);
printf("\n");
}
}

```

## Compilation ,Execution & Testing

```

Session Actions Edit View Help
(kali㉿kali)-[~]
$ vi hdoc24.c

(kali㉿kali)-[~]
$ cc hdoc24.c
hdoc24.c:10:10: function 'consecutiveNumberTriangle'
(kali㉿kali)-[~] unknown type name 'HDOC'
$ ./a.out Day 24 | Page 4
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 1
Enter number of rows: 4
*
* *
* * *
* * * *
hdoc24.c
(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 1
Enter number of rows: 1
*

```

```

(kali㉿kali)-[~] expected '=', '(', '{', '}', 'asm' or '__attribute__'
$ ./a.out Day 24 | Page 4
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 4
1
2 2
3 3 3
4 4 4 4
hdoc24.c
(kali㉿kali)-[~] MENU
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 2
1
2 2

```

```

(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 4
1
1 2
1 2 3
1 2 3 4

(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 3
1
1 2
1 2 3

```

```

(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter number of rows: 4
1
2 3
4 5 6
7 8 9 10

(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter number of rows: 3
1
2 3
4 5 6

```

```

(kali@kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 5
Enter number of rows (1-26): 4
A
A B
A B C
A B C D

```

```

Session Actions Edit View Help
Enter number of rows (1-26): 26
A
A B
A B C
A B C D
A B C D E
A B C D E F
A B C D E F G
A B C D E F G H
A B C D E F G H I
A B C D E F G H I J
A B C D E F G H I J K
A B C D E F G H I J K L
A B C D E F G H I J K L M
A B C D E F G H I J K L M N
A B C D E F G H I J K L M N O
A B C D E F G H I J K L M N O P
A B C D E F G H I J K L M N O P Q
A B C D E F G H I J K L M N O P Q R
A B C D E F G H I J K L M N O P Q R S
A B C D E F G H I J K L M N O P Q R S T
A B C D E F G H I J K L M N O P Q R S T U
A B C D E F G H I J K L M N O P Q R S T U V
A B C D E F G H I J K L M N O P Q R S T U V W
A B C D E F G H I J K L M N O P Q R S T U V W X
A B C D E F G H I J K L M N O P Q R S T U V W X Y
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Increasing natural-number pattern
(kali@kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 5
Enter number of rows (1-26): 27
Invalid input. Enter rows from 1 to 26.

```